Peilin Wang

Jinan, Shandong, China | +86 13361093160 | 202300820012@mail.sdu.edu.cn | Google Scholar

EDUCATION

Shandong University
Bachelor of Science in Statistics Expected Graduation: Jun 2027

- **Overall Average:** 88.75/100 **Ranking**: 16/87 (Top 18.39%)
- **Relevant Coursework:** Functional Analysis; Complex Analysis; Stochastic Processes; PDE; Mathematical Statistics; Bayesian Statistics; Multivariate Statistical Analysis

PAPERS

UNDER REVIEW · ICLR · 2027 · First Author
What to Preserve in Recursive Computation: A Local Predictive Sufficiency Principle
Peilin Wang, Feng Shiyang, Hongfu Gao, Cencheng Zhao, Di Yuan, Hui Chen, Guiguang Ding
Contribution: Proposed the core idea and theoretical framework, developed the main proofs and constrained optimization method, and led the multi-domain validation.

PUBLISHED · International Journal of Financial Studies · JCR Q2 · 2026 · 2nd Student Author
Systemic Financial Risk Spillover Between Traditional-Energy and New-Energy Markets: A Quantile Time–Frequency Network with Link Prediction
Wenxuan Jin, Di Yuan, **Peilin Wang,** Sufang Li
Contribution: Developed the temporal link-prediction module for forecasting emerging directed spillover links and conducted out-of-sample evaluation.

UNDER REVIEW · Energy Economics · JCR Q1 · 2026 · 2nd Student Author
Do Climate Policy Uncertainty and Artificial Intelligence Matter in Energy Markets? A Quantile-Frequency Connectedness Analysis for China and the U.S.
Sufang Li, Mingxu Xiao, Di Yuan, **Peilin Wang**, Kai Du
Contribution: Led the multivariate statistical analysis, using QVAR and forecast-error variance decomposition to characterize state-dependent directional spillovers.

INVENTION PATENTS

CHINESE INVENTION PATENT · Under Substantive Examination · First Inventor
Multi-Source Perception-Based Monitoring and Early-Warning Method and System for Underground Utility Tunnel Operations and Maintenance
Peilin Wang, Shiqi Wang, Xiangcheng Wang, Kehan Chen, Xuan'er Chen, Yaoyang Zhang, Hengjincheng Gong, Yingxiao Yang
Contribution: Conceived the core idea and designed the statistical inference and calibration framework, combining RAG-derived semantic support, surprisal-based anomaly scoring, temporal/multimodal consistency modeling, and task-adaptive calibration of vision-language model predictions.

CHINESE INVENTION PATENT · Under Substantive Examination · 3rd Student Inventor
Edge-Relay-Based Communication Control Method and System for Strong Electromagnetic Interference Environments
Xiangcheng Wang, Yifan Li, Shiqi Wang, **Peilin Wang**, Xuan'er Chen, Shengyuan Wang, Ruizhi Jiao, Yuan Sui
Contribution: Developed the core operations-research formulation using robust optimization, lexicographic multi-objective optimization, and graph-based path optimization to ensure reliable task scheduling and routing under strong electromagnetic interference.

CHINESE INVENTION PATENT · Under Substantive Examination · 2nd Student Inventor
Method and System for Dynamic Enterprise Risk Identification and Prevention
Di Yuan, Shiyang Feng, **Peilin Wang**, Ruomeng Zhang, Xuesheng Chen, Hongfu Gao
Contribution: Designed the core methodology, combining econometric lagged-spillover statistics, weak supervision, HMM smoothing, and heterogeneous-graph modeling to recover reliable dynamic risk-propagation networks from incomplete and noisy enterprise relationships.

CHINESE INVENTION PATENT · Under Substantive Examination · 2nd Student Inventor
Method and System for Communication Networking Relay and Data Transmission in Network-Denied Spaces
Xianfeng Yuan, Hengjincheng Gong, **Peilin Wang**, Junchi Zhang
Contribution: Designed core system algorithms, integrating DQN-based adaptive power control for energy-throughput optimization, heuristic multi-parameter dynamic routing, and TLV-based protocol compression for heterogeneous IoT devices.

PROJECT EXPERIENCE

Intelligent Monitoring System for Nuclear Power Underground Pipe Corridors
Lead Algorithm Engineer Nov.2023 – May.2026
Algorithm Design: Enhanced LLaVA for low-texture, sample-deficient environments with skeleton-aware perception, constrained KG reasoning, and hardware–software communication design.

National Social Science Fund Project: Credit Risk Contagion in New Energy Industry Clusters
Algorithm Architect Oct.2025 – Present
Algorithm Design: Developed a statistical and graph-learning framework for dynamic enterprise risk contagion, integrating financial time-series modeling, semantic-aware heterogeneous graph fusion, and statistically guided generative modeling.

ASC Student Supercomputer Challenge
Core Developer Jan.2026 – Mar.2026
Algorithm Design: Accelerated diffusion-based world-model inference via structured attention and DDIM sampling, achieving a 47.5% speedup.

HONORS

- **Challenge Cup – New Generation Information Technology Track**, National Third Prize; Provincial Special Prize (Top 4 in Province); Team Rank: 1/10
- **RoboCom Robot Developer Competition**, National First Prize, Ranked 7th; Team Rank: 1/3
- **ASC World University Supercomputer Competition,** Global Second Prize; Team Rank: 2/5
- **Mathematical Contest in Modeling (MCM)**, Problem A, Honorable Mention; Team Rank: 1/3
- **China Undergraduate Mathematical Contest in Modeling (CUMCM)**, Problem A,Provincial Second Prize; Team Rank: 1/3
- **Shandong Provincial Artificial Intelligence Competition**, Provincial First Prize; Team Rank: 1/5
- **Second-Class Academic Scholarship**, University-level; Awarded for Two Consecutive Years
- **Discipline-Specific Scholarships**, Innovation Research & Entrepreneurship Practice Scholarship

SKILLS

Languages: CET-6: 518(2025)
Technical Skills: Python, R; PyTorch, TensorFlow, scikit-learn, PyTorch Geometric, Hugging Face; statsmodels; Git, Linux, LaTeX
Interests: Basketball, Badminton, Hip-Hop, and Fitness